Experiment 24: Graphical Model Sampling and Probabilistic Inference

Question

Problem Statement: The clinic from Experiment 23 asks: given a positive X-ray and a persistent cough, how likely is lung disease, and does smoking change the answer? The same medical Bayesian network, or a student-performance network (Difficulty, Intelligence, Grade, SAT, Letter). The objective is to perform exact and approximate inference on a graphical model. Use pgmpy, NumPy, Pandas, NetworkX, Matplotlib.

Tasks:

1. Run exact inference with Variable Elimination for several evidence settings.
2. Generate samples with forward sampling and rejection sampling.
3. Estimate the same queries from the samples using Pandas.
4. Plot estimated probability against sample size to show convergence.
5. Compare exact and approximate answers in a table.

Code

```
"""Experiment 24: Graphical Model Sampling and Probabilistic Inference (medical network)"""
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from pgmpy.factors.discrete import TabularCPD, State
from pgmpy.inference import VariableElimination
from pgmpy.sampling import BayesianModelSampling
try:
    from pgmpy.models import DiscreteBayesianNetwork as BayesianNetwork
except ImportError:
    from pgmpy.models import BayesianNetwork


np.random.seed(0)

# ---------------- Same network as Experiment 23 ----------------
model = BayesianNetwork([("Smoking", "LungDisease"), ("Pollution", "LungDisease"),
                         ("LungDisease", "Cough"), ("LungDisease", "XRay"), ("LungDisease", "Fatigue")])
model.add_cpds(
    TabularCPD("Smoking", 2, [[0.7], [0.3]], state_names={"Smoking": ["no", "yes"]}),
    TabularCPD("Pollution", 2, [[0.8], [0.2]], state_names={"Pollution": ["low", "high"]}),
    TabularCPD("LungDisease", 2, [[0.95, 0.85, 0.70, 0.50], [0.05, 0.15, 0.30, 0.50]],
               evidence=["Smoking", "Pollution"], evidence_card=[2, 2],
               state_names={"LungDisease": ["no", "yes"], "Smoking": ["no", "yes"], "Pollution": ["low",
"high"]}),
    TabularCPD("Cough", 2, [[0.85, 0.20], [0.15, 0.80]], evidence=["LungDisease"], evidence_card=[2],
               state_names={"Cough": ["no", "yes"], "LungDisease": ["no", "yes"]}),
    TabularCPD("XRay", 2, [[0.95, 0.10], [0.05, 0.90]], evidence=["LungDisease"], evidence_card=[2],
               state_names={"XRay": ["negative", "positive"], "LungDisease": ["no", "yes"]}),
    TabularCPD("Fatigue", 2, [[0.80, 0.30], [0.20, 0.70]], evidence=["LungDisease"], evidence_card=[2],
               state_names={"Fatigue": ["no", "yes"], "LungDisease": ["no", "yes"]}))
assert model.check_model()

# ---------------- Task 1: exact inference (Variable Elimination) ----------------
infer = VariableElimination(model)
queries = {
    "P(Lung | no evidence)":                        {},
    "P(Lung | XRay=pos)":                           {"XRay": "positive"},
    "P(Lung | XRay=pos, Cough=yes)":                {"XRay": "positive", "Cough": "yes"},
    "P(Lung | XRay=pos, Cough=yes, Smoking=yes)":    {"XRay": "positive", "Cough": "yes", "Smoking": "yes"},
    "P(Lung | XRay=pos, Cough=yes, Smoking=no)":     {"XRay": "positive", "Cough": "yes", "Smoking": "no"},
}
exact = {}
print("EXACT INFERENCE (Variable Elimination)")
for name, ev in queries.items():
    q = infer.query(["LungDisease"], evidence=ev, show_progress=False)
    exact[name] = float(q.get_value(LungDisease="yes"))
    print(f"  {name:48s} = {exact[name]:.4f}")
print("\n=> Smoking changes the answer: same symptoms, but a smoker has a higher probability of lung
disease.")
```

```
# ---------------- Task 2: forward sampling and rejection sampling ----------------
sampler = BayesianModelSampling(model)
N = 100_000
fwd = sampler.forward_sample(size=N, seed=1, show_progress=False)
print("\nForward samples (first rows):\n", fwd.head())


ev_main = {"XRay": "positive", "Cough": "yes"}
rej = sampler.rejection_sample(evidence=[State(k, v) for k, v in ev_main.items()],
                               size=5000, seed=2, show_progress=False)
print(f"Rejection sampling returned {len(rej)} samples consistent with the evidence {ev_main}")


# ---------------- Task 3: estimate queries from samples with Pandas ----------------
def estimate(df, evidence):
    sub = df
    for var, val in evidence.items():
        sub = sub[sub[var] == val]
    return (sub["LungDisease"] == "yes").mean() if len(sub) > 0 else np.nan, len(sub)


# ---------------- Task 4: convergence with sample size ----------------
sizes = [100, 300, 1000, 3000, 10000, 30000, 100000]
plt.figure(figsize=(9, 5.5))
curve_queries = ["P(Lung | XRay=pos, Cough=yes)", "P(Lung | XRay=pos, Cough=yes, Smoking=yes)",
                 "P(Lung | XRay=pos, Cough=yes, Smoking=no)"]
for name in curve_queries:
    ests = [estimate(fwd.iloc[:s], queries[name])[0] for s in sizes]
    line, = plt.plot(sizes, ests, "o-", label=f"{name} (sampled)")
    plt.axhline(exact[name], color=line.get_color(), ls="--", alpha=0.6)
plt.xscale("log"); plt.xlabel("Number of forward samples"); plt.ylabel("Estimated probability")
plt.title("Convergence of sampled estimates to exact values (dashed)"); plt.legend(fontsize=8);
plt.grid(alpha=0.3); plt.show()

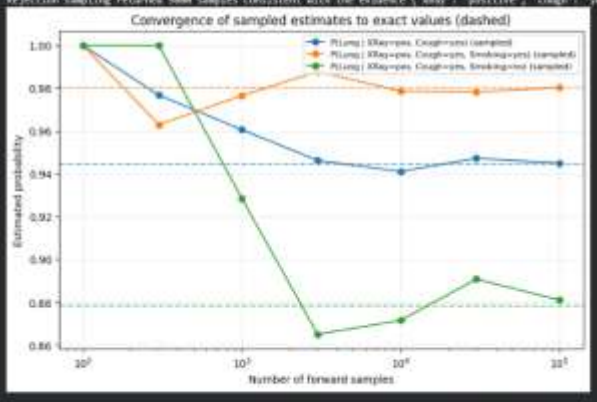

# ---------------- Task 5: exact vs approximate table ----------------
rows = []
for name, ev in queries.items():
    est_f, n_f = estimate(fwd, ev)
    row = {"Query": name, "Exact (VE)": exact[name], "Forward sampling": est_f, "Samples used": n_f}
    if ev == ev_main:                                          # rejection sampling was run for this
evidence
        row["Rejection sampling"] = (rej["LungDisease"] == "yes").mean()
    rows.append(row)
table = pd.DataFrame(rows)
table["Abs error (forward)"] = (table["Exact (VE)"] - table["Forward sampling"]).abs()
pd.set_option("display.width", 220); pd.set_option("display.max_columns", None)
print("\n", table.round(4).to_string(index=False))
```

Output

```
/usr/local/lib/python3.11/dist-packages/pgmpy/estimators/__init__.py:1: FutureWarning: `pgmpy.estimators.StructureScore` is deprecated and will be removed in v1.1.0, use `pgmpy.structure_score` instead.
  from .StructureScore import (
EXACT INFERENCE (variable elimination)
  P(Lung | no evidence)                          = 0.1510
  P(Lung | XRay=pos)                             = 0.7620
  P(Lung | XRay=pos, Cough=yes)                  = 0.9447
  P(Lung | XRay=pos, Cough=yes, Smoking=yes)     = 0.9082
  P(Lung | XRay=pos, Cough=yes, Smoking=no)      = 0.8704

-> Smoking changes the answer: same symptoms, but a smoker has a higher probability of lung disease.

Forward samples (first rows):
   Smoking LungDisease Pollution Cough     XRay Fatigue
0       no         no       low    no  negative      no
1      yes        yes       low   yes  positive     yes
2       no        yes       low   yes  positive     yes
3       no         no       low    no  negative      no
4       no        yes       low   yes  positive     yes
Rejection sampling returned 9000 samples consistent with the evidence {'XRay': 'positive', 'Cough': 'yes'}
```



Query	Exact (VE)	Forward sampling	Samples used	Rejection sampling	Abs error (Forward)
P(Lung | no evidence)	0.1510	0.1506	100000	NaN	0.0004
P(Lung | XRay=pos)	0.7620	0.7611	17799	NaN	0.0009
P(Lung | XRay=pos, Cough=yes)	0.9447	0.9458	11488	0.946	0.0003
P(Lung | XRay=pos, Cough=yes, Smoking=yes)	0.9082	0.9083	7981	NaN	0.0001
P(Lung | XRay=pos, Cough=yes, Smoking=no)	0.8704	0.8661	4007	NaN	0.0043